Q06: What Are Models Computing? Generalization, Memorization, and Emergent Capabilities

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Learning objectives

Students will define generalization relative to a distribution and task, explain why memorization and generalization can coexist, identify what an "emergent capability" plot does and does not prove, and construct a held-out evaluation that rules out at least one shortcut.

1 There is no answer without a distribution

A model is trained on examples and evaluated on cases it did not see. It generalizes when its performance transfers to a specified target distribution. That last phrase matters: a random held-out split, a new wording of the same task, a new composition of known skills, and a new domain are different tests. A score only says what it says about the distribution represented by its test set.

Memorization is also not a single failure label. A model may retain individual training examples, use memorized fragments as part of a broader solution, or exploit accidental correlations that happen to hold in a benchmark. Modern neural networks can fit random labels and still generalize on natural data, a result that makes simplistic capacity explanations inadequate [4]. The practical question is not "memorization or generalization?" but *which information is reused, on which new cases, and with what robustness?*

2 What could a model be computing?

For a next-token model, the immediate computation is a conditional distribution over continuations. To lower loss, its internal representations can encode many regularities: local syntax, factual associations, program patterns, task formats, and possibly procedures that are useful across contexts. We infer a useful computation from controlled behavior, interventions, or mechanistic evidence; we should not infer it solely from a fluent explanation.

Few-shot prompting provides one example of context-sensitive behavior: demonstrations in the prompt can change outputs without changing parameters [1]. A strong test varies the surface form and holds out the combinations that a lookup table or shallow heuristic would fail. A model that succeeds only on familiar templates has shown a narrower capability than one that transfers under these changes.

3 Emergence: observation before explanation

Some studies report sharp-looking gains in particular benchmark measures as scale increases [3]. This observation motivates hypotheses: perhaps a representation, algorithm, or usable context behavior appears after enough training. But sharp plots do not uniquely identify a cause. Thresholded exact-match scores, prompt choices, and nonlinear metrics can make smooth improvement appear discontinuous [2].

A responsible emergence claim therefore reports the underlying continuous metric, model family, prompt protocol, variance, contamination controls, and failures near the claimed threshold. It also compares alternative scales and task formulations. "The curve changed" is a result; "a new internal algorithm appeared" is a stronger mechanistic claim requiring different evidence.

4 A diagnostic ladder for a capability claim

A useful evaluation can become progressively harder while preserving the intended skill. First, test ordinary held-out examples; this checks whether the system simply reproduces its training examples. Next, paraphrase the request or alter irrelevant formatting; this tests sensitivity to surface form. Then hold out combinations of factors, such as an operation paired with an unfamiliar representation. Finally, introduce a distribution shift that changes the causal structure or available tools. A score drop at one rung is evidence about the boundary of the capability, not a reason to erase all results from earlier rungs.

Baselines matter on every rung. A nearest-neighbor retrieval system, a simple program, a frequency heuristic, and a human or expert-designed solver can reveal whether an apparently difficult benchmark has an easy shortcut. Error analysis should be sampled and categorized: was the answer wrong because the model misunderstood the task, used an invalid intermediate step, copied a distractor, lacked a fact, or failed to use an available tool? Without this record, aggregate accuracy can hide the most educational result.

5 What a reproducible report includes

For a module experiment, record the exact data split, prompt template, decoding parameters, model/version, tool permissions, compute budget, scoring code, and exclusions. Keep a small set of adversarial examples separate from development so that they are not repeatedly optimized against. If a benchmark contains public answers or common web text, explicitly discuss contamination risk rather than interpreting a high score as clean evidence of generalization.

The goal is not to demand an impossible proof that a model has never seen anything similar. It is to make competing explanations costly: a claimed rule should survive cases designed to defeat a lookup shortcut, and a claimed robust system should survive ordinary variations in presentation. Reproducibility turns an impressive example into a result another student can challenge.

6 A small evaluation design

Imagine a task in which color and shape jointly determine an answer. Train on most pairs but withhold a diagonal of novel combinations. A memorizing lookup baseline cannot answer the held-out combinations; a rule that has captured the relation can. The included code constructs this split and reports both scores. It does not establish how a language model works. It illustrates the principle that a good test makes intended generalization and shortcut behavior produce different predictions.

7 Takeaway

Generalization is transfer to a declared target distribution, not a vibe conveyed by a demo. Memorization, pattern reuse, and rule-like behavior can coexist. Emergence is a useful empirical description only when the metric and evaluation design are exposed. To learn what a model computes, make competing explanations predict different outcomes and test them.

References

[1] Tom B. Brown et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 2020.

[2] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? *arXiv preprint arXiv:2304.15004*, 2023.

[3] Jason Wei et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.

[4] Chiyuan Zhang et al. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115, 2021.